Gaming the Algorithm: Algorithmic Literacy, Tactical Innovation, and Gig Workers' Collective Resistance in China

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Abstract

Existing scholarship often portrays algorithmic control as a "digital cage" that undermines gig workers' capacity for collective resistance, but few examine how platform algorithms may afford opportunities for resistance that increase gig workers' bargaining leverage. This paper argues that the intensification of real-time performance evaluation and the gamification of labor have produced a new work regime of hegemonic algorithmic control, which, paradoxically, enables gig workers to develop tactical innovations that outmaneuver their opponents during strikes. Based on four months of ethnographic observation and 65 in-depth interviews in Guangdong, China, I trace the history of labor actions staged by food-delivery and ride-hailing workers in two cities in response to wage cuts. I find that the gamification of the platform inadvertently created a new class of workers – "top players", who possess "algorithmic literacy", the digital skill set required to interpret and game the algorithm. Despite their inability to directly access the algorithm, these top players engaged in "tactical innovation" to increase their bargaining leverage through three mechanisms: algorithmic manipulation (shaping inputs to influence outputs), algorithmic deception (rule-bending tactics that exploit loopholes and avoid detection), and algorithmic non-correction (withholding corrective labor to inflict additional disruption). Together, these mechanisms explain how tactics that emerge from the everyday labor process can be scaled up into tactics of coordinated disruption during strikes. The paper demonstrates how hegemonic algorithmic control can backfire, identifies a new kind of structural power, and advances a labor-process-centered framework for understanding collective action in the platform economy.

Keywords: Algorithmic Control, Collective Action, Labor Movement, Platform Work, Hidden Transcript, China

Introduction

The platform economy has experienced explosive growth, reshaping the structure of work and employment relations worldwide over the past decade. The size of the online platform workforce exceeds 435 million, accounting for at least 12 percent of all workers worldwide (Datta et al. 2023). Platforms are "multisided digital frameworks that shape the terms by which various

actors interact based on the application of algorithms" (Lei 2021). When used to facilitate economic transactions, algorithms have become the "de facto manager" in charge of task assignment, pricing, work instruction, and performance evaluation, mediating the relationship between different clients, merchants, and workers (Cameron 2024). As breakthroughs in AI diffuse across the service industry, such as food delivery and ride-hailing, scholars have examined both the promises and the challenges of gig work (Vallas and Schor 2020). Despite widely championed discourse around entrepreneurship and employment opportunity for marginalized groups, algorithmic management in the labor process has fueled the proliferation of precarious labor (Griesbach 2025), marked by limited autonomy, downward pressure on wages, heightened exposure to risk, and intensified exploitation (Lata, Burdon and Reddel 2023; Rosenblat 2018). Therefore, existing literature often depicts platforms as an invisible yet all-encompassing "digital cage" (Kellogg, Valentine and Christin 2020; Lei 2023; Rahman 2021). The rise of algorithmic management has been compared to a form of "digital Taylorism" (Shibata 2020) that propels labor relations back toward the logics of early industrial capitalism (Ravenelle 2019). To emphasize these dynamics, scholars have also invoked terms such as "algorithmic tyranny" (Lehdonvirta 2018) and "algorithmic despotism" (Griesbach et al. 2019), highlighting the opacity of evaluation systems, the arbitrariness of managerial decision-making, and the steady erosion of workers' capacity for collective resistance (Cameron and Rahman 2022; Tassinari and Maccarrone 2020).

Contrary to the bleak predictions of the "digital cage" perspective, collective action among gig workers has been reported worldwide, even in authoritarian regimes conventionally thought to be hostile to labor movements. Between 2017 and 2020, scholars recorded 1,271 protest events by platform workers across various sectors (Bessa et al. 2022; Umney et al. 2024). In China, in particular, where the political opportunity structure is tightly constrained and collective action is

often deemed unlikely (Hu 2021; Pun 2021), various forms of collective action, including large-scale strikes, have nevertheless taken place among gig workers in response to wage cuts and unfair managerial practices (Huang 2025; Lei 2021; Liu and Friedman 2021; Zhao and Luo 2023). These collective actions in the platform economy have constituted an empirical puzzle (Della Porta, Chesta and Cini 2023; Joyce, Stuart and Forde 2023): why, and through what mechanisms, do gig workers manage to stage collective actions against all odds?

Building on the insights from social movement studies and the sociology of work literature, this article seeks to unpack this puzzle by analyzing the tactical innovations of "gaming the algorithm" that enable collective resistance under algorithmic control. McAdam (1983) introduced the concept of tactical innovation to describe how challengers creatively devise new tactics or techniques that offset their initial powerlessness and "temporarily increase protesters' bargaining leverage," much like moves in a strategic game of chess. According to Tarrow (1998), tactical innovation emerges through the dynamic interaction between activists and their adversaries, a dialectical process evident in both industrial relations and social movements. Successful challengers of the status quo must therefore remain flexible on tactical choices (Snow et al. 2018), outmaneuver their opponents by continually adapting, recombining, and innovating tactics—today often by leveraging digital infrastructures (Earl 2018).

While scholars of social movements have long emphasized the emergence and diffusion of tactical innovations (Edwards 2014; Wang and Soule 2016), labor researchers have developed complementary frameworks to understand the bases of workers' bargaining leverage that these tactics can elicit. The power resource approach, in particular, identifies structural power as one of the primary sources of workers' leverage (Rhomberg and Lopez 2021), which is understood as the

bargaining advantage of an individual worker due to tight labor market conditions or due to their occupation in a strategic location within a key industrial sector (Wright 2000). This form of power provides the most immediate context for understanding workers' capacity to disrupt production, a capacity that Tarrow (1998:98) describes as "the strongest weapon of social movements," offering weaker actors leverage to challenge far more powerful employers.

It is difficult for labor movements to develop new tactics to gain leverage (Voss 1996), and this challenge is even more pronounced in the platform economy. Platform companies exercise what I term "hegemonic algorithmic control": managerial strategies that secure worker compliance not through coercion but through the manufacture of consent (Burawoy 1985; Cameron 2024). These algorithmic platforms achieve hegemonic algorithmic control by combining data-driven performance evaluation, selective rewards, behavioral nudges, and game-like design features, echoing Burawoy's (1979) observation of "making out" games under piece-rate systems. Yet, scholars in the labor process tradition have long emphasized that managerial control and worker resistance are dialectically intertwined at the point of production (Braverman 1974; Burawoy 1979; Edwards 1982; Rhomberg and Lopez 2021; Schmalz, Ludwig and Webster 2018). Bringing these two stream of studies together (Silver and Karatasli 2014), this paper examines how algorithmic control can inadvertently create new openings for gig workers to devise tactics and cultivate collective resistance in their everyday interactions with the algorithm.

Based on four months of ethnographic observation and sixty-five in-depth interviews with various gig workers in China, I traced the history of collective resistance among food-delivery and ride-hailing platform workers in Dongguan and Shanwei since 2018, when these services first took root in the two cities. I find that hegemonic algorithmic control backfires as workers collectively

game the algorithm. Ironically, the gamification of work has inadvertently produced a new group of workers: top players. These workers develop a high level of algorithmic literacy, defined as the ability and interpretive skill set that gig workers accumulate through repeated human-algorithm interactions, enabling them to decipher algorithmic decisions, strategically influence algorithmic outputs, and exploit loopholes in the system. By sharing, testing, and refining repertoires for gaming the algorithm, top players collectively transform everyday know-how into bargaining leverage during strikes and coordinated logoffs.

As illustrated in Figure 1, "gaming the algorithm" constitutes a form of tactical innovation that emerged from gig workers' everyday labor process and was deployed during collective actions, allowing workers to momentarily subvert managerial control and convert algorithmic knowledge into structural power through three mechanisms. First, top players manipulate the algorithm to secure material advantages and favorable performance metrics. Second, they creatively deceive the algorithms, bending rules while avoiding sanctions. Third, they engaged in algorithmic non-corrections, withholding their invisible labor to correct and override the malfunctioning algorithm, thereby causing additional disruption, raising their replacement cost, and strengthening their bargaining leverage. These tactics are collectively refined and shared through peer networks, along with moral narratives that justify their deployment for various purposes, constituting a "hidden transcript" among gig workers and top players. The tactics of gaming the algorithm can be particularly disruptive during strikes when used collectively, forcing management to make concessions even when traditional sources of worker power remain weak in the fragmented, precarious gig economy.

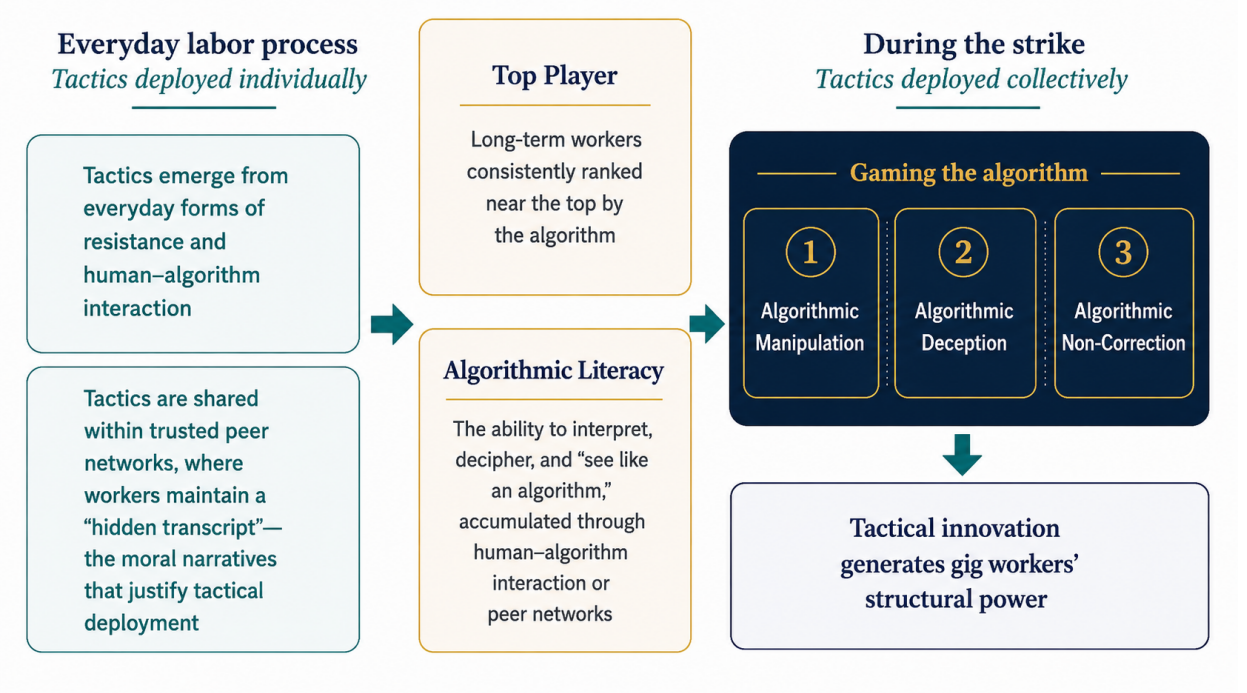


Figure 1. A Conceptual Map on Gaming the Algorithm as Tactical Innovation.

This paper shows that hegemonic algorithmic control is not an unbreakable "iron cage" but a contradictory regime that can backfire. As platforms intensified performance evaluation and reduced piece rates, they generated widespread grievances and weakened workers' willingness to keep playing the game. At the same time, the everyday labor process cultivates algorithmic literacy and tactical repertoires among top players, allowing them to convert platform knowledge into structural power during strikes and coordinated logoffs. The paper first theorizes hegemonic algorithmic control and resistance, then examines the formation of algorithmic literacy, and finally analyzes how workers transform this literacy into tactical innovation and collective leverage.

Theoretical Framework

The Spectrum of Algorithmic Control: From Despotic to Hegemonic

In the platform economy, algorithms coordinate labor processes and mediate relationships among business owners, customers, and gig workers (Cameron 2024; Duggan et al. 2020). Lee et al. (2015) first refer to such development as "algorithmic management": the delegation of core managerial functions to software algorithms and the institutional arrangements that enable their operation in practice. The introduction of AI-powered digital infrastructure on platforms has, nevertheless, resulted in tighter labor control and diminishing prospect of labor resistance (Rahman 2024). Existing literature has characterized these emerging work regimes along a continuum from despotic to hegemonic forms of algorithmic control. Griesbach et al. (2019) describe "algorithmic despotisms" as a mode of management in which platform companies exercise coercive authority over workers through arbitrary instructions and undisputable systems of rewards and punishments. For instance, platforms such as Instacart set up task-allocation algorithms that effectively require workers to maintain long, fixed hours as a precondition for receiving orders, while penalizing those who reject their task assignments. Likewise, many platforms often deploy stringent and yet opaque performance-evaluation algorithms to determine pay and access to future work opportunities (Rahman 2024; Zhao and Luo 2023). Under algorithmic despotism, gig workers experience pervasive grievances: the rules of algorithms are subject to unilateral changes without any channel of redress, and workers lack protections against injury as well as unjust dismissal (Lei 2021). In this sense, the workplace under algorithmic control has become, as Ravenelle (2019) puts it, a "movement forward to the past."

Yet, as Burawoy (1979) it reminds us, labor control does not rely on coercion alone. It is also achieved through hegemony, namely, the manufacture of consent, where exploitation is transformed into voluntary participation as workers internalize the pursuit of employer-defined goals. Platform companies have extended this logic through gamification, "the use of game

elements—point scoring, levels, leaderboards, competition, and measurable evidence of accomplishment—in non-game contexts" (Mason 2018). Under mounting pressure to simultaneously sustain efficiency and profitability (Shestakofsky 2017), platforms deploy gamified incentive systems that blend performance monitoring with emotional engagement, offering workers a false sense of choice and progress within a tightly managed labor process (Cameron 2024). In food delivery platforms, for instance, companies design ranking systems modeled after popular mobile games, classifying riders into hierarchical status and titles such as Bronze, Silver, Gold, and King, and workers compete in monthly challenges for cash bonuses, smartphones, or preferential access to orders (Bonini and Treré 2024). Likewise, Uber and Lyft drivers often extend working hours or perform unpaid labor to recover unfavorable ratings and maintain a "five-star" status (Mason 2018). Such gamified systems not only channel workers' motivation into intensified competition but also exacerbate informational asymmetries: algorithms monitor every move, granting management a panoptic view of performance while keeping workers in the dark, often nudging them toward low-paid tasks (Lata, Burdon and Reddel 2023). This shift in algorithmic management exemplifies a hegemonic labor regime (Burawoy 1985), where control operates less through coercion than through the subtle internalization of algorithmic norms and metrics of success—what might even be described as a form of digital biopolitics (Gregory and Sadowski 2021).

Studies on the spectrum of algorithmic control offer a bleak portrayal of workers' rights under platform capitalism. Gig workers trapped in the "invisible cage" by opaque algorithms lament that "accept or do not, you won't win the fight." (Rahman 2021). Even when workers attempt to circumvent managerial directives by manipulating algorithmic input, such actions reinforce consent to their own self-exploitation (Cameron 2024). However, the one-sided accounts

that portray algorithms as an all-powerful apparatus risk obscuring workers' agency and treating labor resistance as nothing more than a puzzle (Joyce, Stuart and Forde 2023; Vallas and Schor 2020; Woodcock 2021). Reexamining the labor process through the metaphor of the game illuminates how even hegemonic control can backfire, contradicting the image of this all-encompassing digital cage. For example, Burawoy (1979) original observation shows that gamification ceases to be effective when the game's outcomes are no longer attractive or attainable. Game studies also remind us that wherever there is a game, there is also "cheating": players routinely bend or subvert rules to gain advantage in ways that designers neither anticipate nor fully suppress (Consalvo 2009). Moreover, science and technology studies have demonstrated that workers' and users' efforts to "game" algorithmic systems are not merely deviant or deceptive practices, but situated responses to measurement, ranking, and platform power through which they attempt to regain agency under opaque and unevenly enforced rules (Ziewitz 2019). Therefore, one way to move beyond is to investigate how gig workers interact with algorithms in unintended ways and whether these acts can generate new sources of bargaining power, as discussed in the next section.

The Algorithm of Resistance

An emerging literature on the "algorithms of resistance" offers an alternative account that highlights how workers appropriate, manipulate, or repurpose algorithmic systems to their advantage through everyday forms of resistance such as collective log-offs and coordinated order refusal (Bonini and Treré 2024). Building on the concept of "algorithmic affordances", the actionable possibilities enabled by an algorithm's design and operation, this line of research shows that workers need not seize formal control over algorithms to exploit their unintended uses

(Ettlinger 2018; Kear 2022). Across the platform economy, workers and activists develop tactics to "game" algorithmic systems: leveraging search algorithms to boost visibility or correct biased results (Velkova and Kaun 2021), using multiple accounts, apps, or bots to circumvent unjust account suspensions (Grohmann et al. 2022), screening customers or reshuffling orders to improve earnings and metrics in Uber (Vasudevan and Chan 2022), and engaging in mutual aid to enhance performance evaluation results despite explicit prohibitions in the food-delivery platforms (Zhao and Luo 2023).

A related strand emphasizes the "hidden transcript" and the solidarity-building spaces through which workers contest platform power (Tassinari and Maccarrone 2020). These spaces not only function as communication channels for critiquing unfair platform practices (Zhou and Pun 2024) but also serve as informal learning environments where workers share knowledge of algorithmic rules and develop techniques to optimize their earnings (Anwar and Graham 2020). Moreover, Kear (2022) draws on the moral economy framework to explain how individualized algorithmic resistance can scale into collective action. In the platform economy, gig workers use social media to circulate moralized narratives of algorithmic injustice, and, when conditions permit, mobilize existing online networks to coordinate unionization efforts or even global social movements (Pei 2025; Souza 2024; Woodcock 2021).

Tactical Innovation, Algorithmic Literacy, and the Making of Top Players

While the "algorithm of resistance" literature richly documents how gig workers use everyday forms of resistance to contest platform power, it rarely explains how these dispersed practices translate into concrete bargaining leverage. To address this gap, I conceptualize "gaming the algorithm" as a contemporary form of tactical innovation in algorithmically mediated

workplaces. In classic social movement accounts, tactical innovation refers to the way challengers develop new repertoires of contention in response to shifting opportunities and constraints (Tilly 2003; Wang and Soule 2016). Originating from activist networks and diffusing through social media (Andrews and Biggs 2006; McAdam 1983; Morris 1984), innovative tactics temporarily increase protesters' bargaining leverage against powerful opponents (McAdam 1983; Taylor and Van Dyke 2004), and thus, deploying effective tactics is crucial for the success of labor movements (McAdam, McCarthy and Zald 1996; Voss 1996). Similarly, gig workers' practices of algorithmic manipulation, deception, and non-correction emerge first as improvised solutions or everyday forms of resistance in a gamified labor process, often deployed by top players. Over time, these individualized playbooks diffuse through peer networks and are ultimately transformed into tactics that can be collectively coordinated and strategically deployed during strikes. In this sense, "gaming the algorithm" serves as a digital-era analog to tactical innovation. It transforms mundane, individualized resistance into coordinated forms of disruption that enhance workers' bargaining leverage, or "structural power" (Wright 2000), against platform management during strikes or collective logoffs.

The capacity to "game the algorithm", as a tactic of resistance, is grounded in gig workers' algorithmic literacy. This concept draws on parallels from financial credit-scoring contexts, where marginalized subjects cultivate "financial literacy" to contest their subordination within financial systems (Fourcade and Healy 2024). For gig workers in particular, a major obstacle to turning the algorithm to their advantage lies in its opacity, which, as Rahman (2021) argues, is intentionally designed to prevent workers from "gaming the system." In response, gig workers have begun forming self-organized learning communities to enhance algorithmic literacy, as proficiency with the algorithmic decision-making processes becomes a crucial foundation for collective resistance.

Through these efforts, workers exchange knowledge and develop practical repertoires for navigating algorithmic control. For example, Uber drivers share recorded videos exposing the "scams" behind surge-pricing mechanisms and Upwork freelancers host offline workshops and offer peer training through social media groups to educate newcomers about platform rules and loopholes (Anwar and Graham 2020).

However, algorithmic literacy, as a skill set, also serves as a stratifying mechanism in the platform economy. Tactical innovations among gig workers remain relatively scarce, in part because of the emerging digital divide that has left already marginalized groups even more vulnerable (Gran, Booth and Bucher 2021), while gig workers in relatively advantaged social positions, such as long-term participants in the industry or workers with connections to management, often possess greater familiarity with how algorithms operate, which experience enables them to "game the algorithm" more effectively to secure better pay and working conditions. Drawing on the game metaphor embedded in workers' own vernacular of "gaming the algorithm", I use the term "top players" to describe workers who are often ranked the highest by the algorithm's evaluation of their performance. In online gaming communities, they are those who acquire specialized knowledge to gain an advantage through creative manipulation or "cheating" (Consalvo 2009). Moreover, this distinction among gig workers resonates with Bourdieu (1987) the emphasis on how accumulated forms of capital – and, in the context of the platform economy, algorithmic literacy accumulated through repeated interaction with the algorithm – differentiate these workers from their peers. As Schor, Tirrell and Vallas (2024) noted, gig workers' background factors shape their differential exposure to risks and heterogeneous working experiences. The concept of "top players" joins this line of studies by focusing on how heterogeneous access to algorithmic literacy, as an expression of pre-existing social inequality, generates a new hierarchy

among gig workers and transforms some of them into semi-skilled actors capable of converting digital knowledge into either material gains or bargaining leverage. In short, this paper addresses how platform workers secure bargaining leverage under algorithmic control and argues that "gaming the algorithm," as a form of tactical innovation, enhances their structural power. The following section outlines the methodological approach and examines these dynamics empirically, tracing how gig workers game the algorithms for collective resistance.

Methodology

This study draws on four months of ethnographic fieldwork and sixty-five in-depth interviews with gig workers in Guangdong Province, China. China currently hosts the world's largest online food-delivery and ride-hailing markets[1]. Its rapid expansion has been accompanied by escalating labor unrest: at least 152 large-scale strikes and protests by gig workers have been reported between 2017 and 2024[2]. Although such activism has faced intensified repression, such as the detention of autonomous union leaders in the 2021 crackdown on Beijing platform couriers' alliance[3], gig workers continue to engage in various forms of collective resistance. Instead of adapting to repression by keeping a low profile like many NGO leaders and activists in China (O'Brien 2022), they persist in highly visible and confrontational strikes. Guangdong serves as an ideal site for this study because it is designated as the policy pilot zone for China's Digital Bay Area project, which channels significant investment into digital infrastructure and platform economy[4]. The province's pivotal role in the platform economy makes it a critical setting for examining the dialectics of algorithmic control and collective resistance it provokes.

The fieldwork and interviews were carried out between 2023 and 2025 and focused on ride-hailing and food-delivery workers, two major sectors of China's urban platform workforce. I selected Shanwei as the first field site using the negative case method (Emigh 1997). Given the atomization of gig work and the intensification of state repression, large-scale gig worker strikes are theoretically unlikely. Shanwei, therefore, offered a striking deviant case: despite these

[1] https://finance.yahoo.com/news/china-online-food-delivery-market-120800151.html. Accessed Oct 30, 2025. https://www.sixthtone.com/news/1015043. Accessed Oct 30, 2025.
[2] https://maps.clb.org.hk. Accessed May 4, 2024.
[3] https://labornotes.org/2021/04/china-leader-delivery-riders-alliance-detained-solidarity-movement-repressed. Accessed May 4, 2024.
[4] https://www.scmp.com/economy/china-economy/article/3242471/chinas-digitalisation-push-seeks-digital-energy-spark-vital-tech-economic-growth-greater-bay-area. Accessed Oct 30, 2025.

constraints, it became the site of China's largest gig worker strike (Han and Song 2024). Studying this case allowed me to identify new mechanisms that would be difficult to discern in more routine settings (Seawright and Gerring 2008). In the next stage, I conducted subsequent interviews in Dongguan, a more typical Pearl River Delta city with a predominantly migrant workforce, to assess whether these mechanisms recurred beyond the original case (Small 2009). When similar tactics emerge across cases with different city-level socioeconomic and demographic conditions, the shared algorithm-mediated platform work regime becomes a more plausible key factor in gig workers' collective actions (Ragin 2014; Skocpol and Somers 1980). Platform companies first began operations in Dongguan (around 2016) and later expanded to Shanwei two years later. During the study period, workers in both sites and across platforms experienced similar cuts to piece rates and rollbacks of benefits.

To trace the history and dynamics of gig worker resistance, I conducted 65 semi-structured interviews with workers and platform intermediaries, each lasting an average of one hour and forty-five minutes. Interviews were usually conducted during workers' idle periods, such as after the lunch rush or late at night after work. During recruitment, I introduced myself as a researcher conducting a university-based social science project on the platform economy and informed participants that they would receive 50 yuan (approximately US$7) in compensation for their time. In addition to interviews, I conducted ethnographic observation in workers' everyday social and work settings in both cities. After building rapport with informants, I joined workers at late-night barbecues, informal gatherings, waiting sites, and other spaces where they exchanged information, discussed platform rules, and reflected on collective action. When feasible, I also accompanied food-delivery couriers during their work routines, ∂observing how they navigated order assignments, delivery deadlines, customer interactions, and algorithmic penalties in real time.

These observations provided access to the informal interactions, tacit knowledge, and situated practices through which workers interpreted, negotiated, and resisted algorithmic control.

I used three recruitment strategies to capture variation across locations, platform status, and work histories. First, I placed orders through delivery and ride-hailing platforms and interviewed workers who accepted them, varying the timing and location of orders to reach workers across different shifts and neighborhoods. Second, I conducted site-based recruitment in locations where gig workers waited for assignments, including commercial districts, restaurant clusters, and transportation hubs; this strategy helped include workers who might be underrepresented in platform-mediated recruitment because of low rankings or limited order access. Third, I used snowball sampling to reach long-term workers who had joined the platforms near their local establishments and could provide retrospective accounts of platform changes and earlier labor actions. The sample includes 31 food-delivery couriers working for Meituan and Ele.me, China's two largest delivery platforms, and 34 ride-hailing drivers, most of whom primarily used Didi Chuxing while also maintaining accounts on multiple apps. All but two informants were men between the ages of 19 and 45, with one female informant in each sector. In addition to rank-and-file workers, I interviewed six informants in managerial positions, ranging from junior team leaders to upper-level station managers. Most respondents had completed junior high or high school; none had graduated from a four-year university, though several held associate degrees[5]. To confirm my findings, I also conducted several supplementary interviews with workers in other cities (i.e., Shenzhen) and cities outside Guangdong (i.e., Wuhan).

[5] To protect informants' anonymity, I report aggregated sample characteristics in Appendix Table A1 and use pseudonyms throughout. Interview excerpts are identified by anonymized codes indicating field site, sector, and interview number: DGF refers to a Dongguan food-delivery worker, DGR to a Dongguan ride-hailing driver, SWF to a Shanwei food-delivery worker, and SWR to a Shanwei ride-hailing driver.

Findings

Since platforms entered my field sites in Shanwei and Dongguan, workers in both cities have repeatedly mobilized in response to wage cuts and deteriorating conditions. In Shanwei, food-delivery couriers staged major strikes in 2020, 2021, and 2023; the 2023 stoppage lasted more than twelve days and reportedly involved over 800 couriers citywide. Ride-hailing drivers also organized a coordinated "log-off" strike in 2018. In Dongguan, food-delivery workers launched strikes in 2018 and again in 2023, though these actions were generally smaller and more fragmented. In line with existing research (Liu and Friedman 2021), these findings suggest that, overall, collective actions in China's platform economy are far more frequent than previously reported by social media and labor NGOs.

A key reason for this undercount is that much of the resistance is not publicly visible to outsiders. Instead, gig workers often contest platform control inside the labor process through what I call "gaming the algorithm"—a form of tactical innovation rooted in everyday worker–algorithm interaction and diffused through trusted peer networks. By "gaming, I mean that workers accept the platform's gamified rules but repurpose them for their own advantage and "cheat" in small ways. These practices usually begin as individualized, routine "tricks" for getting by, but can "scale up" later when deployed collectively to enhance bargaining leverage during strikes. The sections that follow identify three mechanisms through which algorithmic gaming generates structural power, namely, algorithmic manipulation, algorithmic deception, and algorithmic non-correction. I organize the discussion of each mechanism in two steps: first, how top players learn, test, and refine the tactic through everyday work; and second, how workers redeploy the tactic collectively as a disruptive strategy during moments of open contention. I conclude the section by identifying

the conditions that enable "gaming the algorithm", namely, the emergence of top players and the accumulation of algorithmic literacy.

Gaming the Algorithm as Tactical Innovations

Algorithmic manipulation

Algorithmic manipulation refers to workers' collaborative efforts to subtly alter the algorithm's inputs to obtain desirable outputs. This is a low-risk tactic because it operates within the platform's existing rules, but artificially alters inputs to influence the algorithm's decision-making process. A typical example in the everyday labor process is that food-delivery gig workers collectively delay accepting orders to trigger the platform's surge-pricing algorithm and raise the per-order rate. Among "experienced workers" (熟手), this practice is governed by a tacit norm of "knowing the rules" (懂规矩): when an order appears at an unacceptably low rate, veteran players avoid competing for it and instead wait together until the system automatically increases the piece rate, at which point they take the order[6]. In effect, collective non-acceptance of orders turns the time an order remains unclaimed into a manipulable input, allowing workers to artificially simulate labor scarcity and push the pricing algorithm toward higher pay. Crucially, the tactic only works as a collective practice and depends on shared algorithmic literacy about how and when the platform adjusts rates. Veteran couriers in both cities repeatedly blamed the influx of "rookies" (newcomers) for unraveling this practice: lacking insider knowledge of the pricing rule, newcomers also broke group solidarity by quickly grabbing low-paying orders for immediate individual gain, preventing the coordinated delay and pushing piece rates downward[7]. What many

[6] Interview DGF2408.
[7] Interview SWF2309.

seasoned workers referred to as the "rookie problem" was, in effect, a classic "collective action problem" (Olson 1965), namely, the problem of free-riders who reap short-term gains at the expense of collective benefits. An isolated worker's delayed response would then be dismissed as noise in the system. Only synchronized delays by a sufficiently large group can shift the algorithm's input data and trigger higher pay rates.

How does algorithmic manipulation, then, become a disruptive collective tactic of resistance? During strikes, elite gig workers repurpose everyday algorithmic resistance, such as delaying order acceptance or gaming performance metrics, into collective actions that strategically disrupt platform operations. In Dongguan's 2018 food-delivery worker strike, for example, a small group of top players placed small orders themselves in geographically marginal areas, creating an artificial surge of demand in those zones. Meituan's order assignment algorithm, misled by this distorted input, dispatched nearly all available couriers at that time to the periphery zone—only for them to be recruited to participate in the strike upon arrival[8]. This maneuver effectively hollowed out the local station's workforce, exacerbating labor shortages and inflicting measurable financial losses on station managers, who were then compelled to enter negotiations just after the first day of the strike. Similar tactics were also deployed in the ride-hailing platform. In the midst of a log-off campaign to protest wage cuts in Shanwei, Zhung, a Didi driver on strike, recalled placing "fake orders" to help recruit gig workers:

> We just wanna see if anyone was still online, so we place "fake orders" to find out. For example, I placed a ride request myself to see if the driver will take it or not. If that driver

[8] Interview DGF2408; Interview DGF2401.

> did accept it, call that person right away and tell them [something like] "we are on strike and you better not working and log off immediately"[9].

In both cases, the algorithm was actively manipulated to misallocate labor, catalyze collective disruption, and thus increase workers' bargaining leverage.

Algorithmic deception

The second mechanism is deceiving the algorithm, that is, collectively bending the rules to exploit system loopholes while avoiding detection or punishment. Gig workers often frame this as "finding bugs," invoking the language of video games, where cheating is risky but potentially rewarding (Consalvo 2009). For instance, gig workers often game the geolocation tracking system to simulate availability without actually working. Since staying online is crucial for securing high-paying orders and maintaining reputation scores (Wood et al. 2019), experienced food-delivery couriers would ride to peripheral areas with low demand and disable their GPS, tricking the system into logging idle availability while they took some breaks (or did an interview with me). This tactic, dubbed "idling" (挂单), which got its name from the popular game genre "idle game" (挂机游戏), allowed workers to appear active while effectively pausing work[10]. Top players also deceive the algorithm to avoid unreasonable customers. Chang, a seasoned Meituan courier, discovered that the system marked an order as delivered as long as it was within a 200-meter radius of the destination. When customers refused to answer calls or made additional requests, such as buying

[9] Interview SWF2308. The informant had worked on both food-delivery and ride-hailing platforms.
[10] Interview SWF2305. Interview SWF2301.

cigarettes without pay, he and others would simply drop the food nearby and register the order as complete[11]. To avoid platform fines, elite workers have to constantly experiment with the algorithm's detection threshold, going back and forth to refine their tactics to stay just within acceptable bounds[12]. Ride-hailing drivers employed similar tactics. Su, a Shanwei Didi driver, recounted how he parked near, but not directly at, the pickup point when facing rude customers. He marked himself as "arrived," waited five minutes, and then collected a cancellation fee while avoiding triggering detection[13]. Others described a "free bonus meal" (加餐): if a driver was denied access to a residential compound and could not reach the customer after thirty minutes, they simply kept the food without panelty, as long as the algorithm tracked that they had made attempts[14].

The most high-risk tactic was taking "private orders" for ride-hailing drivers. This involved bypassing the platform entirely to negotiate rides directly with customers, common in popular local tourist attractions, where both parties wanted to avoid long queues and high platform commissions. It's not surprising that this was strictly prohibited, given that platform companies had previously threatened account deactivation by tracking geolocation trails for ride-like patterns, and even penalized drivers for having accounts with rival companies. Hui, a veteran Didi driver, was locked out for a month after the system flagged a suspected private ride. However, knowing the playbook well, the top player persistently called customer service every day for a whole month, until he found a negotiable agent who offered to take a bribe for reinstating his account[15].

[11] Interview DGF2410.
[12] Interview SWF2309.
[13] Interview SWR2309.
[14] Interview DGF2410.
[15] Interview SWR2303.

During the 2023 Shanwei city-wide food-delivery strike, gig workers weaponized algorithmic deception, turning it into a tactic of disruption. On the first day, unsure of how long the strike would last, Wong cautiously joined by accepting a single order but deliberately withheld pressing "delivery complete," keeping his status active without fulfilling the task. When the management demanded that all couriers log on or face permanent deactivation, he employed the "away from order" trick, masking his inactivity as availability by deceiving how the algorithm interpreted his geolocation[16]. Deployed at scale, this tactic created an algorithmic illusion: many workers appeared online, but few completed deliveries, leading to a backlog of unfinished orders. Enraged but lacking proof of rule violations, the managers retaliated on day two by randomly deactivating accounts, as well as manually identifying and relocating "active" couriers to an isolated zone without incoming orders[17]. Nevertheless, as with any collective action, there was safety in numbers. The deception worked precisely because it was deployed en masse, shielding individuals under the veil of coordinated ambiguity. Arbitrary punishment only deepened worker solidarity and undermined the threat's credibility. Likewise, other tactics of algorithmic deception, such as "private orders," went rampant during the strikes among gig workers in ride-hailing platforms[18]. In this context, algorithmic deception was converted into a tactic during the strike, giving them a temporary bargaining advantage by temporarily outmaneuvering the management.

Algorithmic non-correction

The third mechanism is algorithmic non-correction: top players strategically withhold the invisible corrective labor that normally keeps a flawed algorithmic system running. By refusing to

[16] Interview SWF2302.
[17] Interview SWF2309.
[18] Interview SWR2303.

correct the algorithm when it malfunctions, they deliberately allow operational chaos to escalate, generating disruption that compels managerial concession. It inflicts additional disruption beyond simply withdrawing labor, because without the hidden expertise that keeps the algorithm running in normal order, replacement workers are far less able to compensate for the labor shortage. This tactic grows directly out of the everyday labor process. Under normal conditions, top players constantly use their algorithmic literacy and on-the-ground expertise to bypass the malfunctioning algorithm. During peak hours or severe weather, the task-assignment algorithm struggles to cope with sudden spikes in demand. Incoming orders overwhelm the system, causing it to misallocate tasks, sometimes assigning too many orders to a single courier, other times delaying assignment until orders are close to timing out. At the same time, route-planning algorithms can no longer optimize rapidly accumulating deliveries across flooded streets, traffic jams, or gated compounds. These malfunctions are especially punishing for "rookies." Without deep familiarity with local geography or the tacit skill required to reorganize delivery routes manually, inexperienced workers frequently stumble. Their delays cascade into backlogs and plummeting performance scores, amplifying the consequences of algorithmic failure. Managers are not unaware of this algorithmic vulnerability that can lead to severe financial loss and plummeting performance evaluation results[19]. Ho, a former station manager, explained how managers are in desperate need of gig workers who can clear the backlogs during algorithmic malfunction:

> During peak hours, I had to sit at the backstage of the platform and look at the control panel, where it tracked all existing orders and the names of every single food-delivery courier. If there was heavy rain and people did not want to deliver, many orders would go overtime and be flagged as red. After the orders went red, the backstage manager must watch the

[19] Interview DGF2410.

> manpower data. Let's say this happened at 7pm, during dinner hours. There were folks who worked earlier shifts and others who usually work late at night – they do not have to log on [at the moment]. But you had to ask these people to help out now ..... Usually, as station managers, we had to tell team leaders to call their best men to log on. [20]

Top players are always among the first workers called upon to clear backlogs and curb financial losses. From their perspective, this is understood as "rescue orders," as Fu, a veteran courier, explained:

> The station manager knows veteran couriers like us can basically respond at any time. No matter if there are any bad orders, rotten orders, or overtime orders, we can come to the rescue for you [the managers]. You can call me even if I'm eating—I'll put down my chopsticks, run the order for you, and come back to finish eating. We go that far ..... We are the ones they [the managers] use for emergency rescues.[21]

Top players' unique leverage lies in their capacity to correct the malfunctioning algorithm. When a typhoon hit Shanwei, I was struck not by workers hiding indoors but by top players racing through the city—focused and even exhilarated. Unlike baffled newcomers, they spoke proudly of "rescuing orders," treating algorithmic failure as a challenge to beat the game rather than a hindrance to be avoided. They know not only how to game the algorithm when it functions properly, but, crucially, when not to rely on it. When the algorithm fails to adapt to real-life disruptions, top players manually override the algorithm, shortcut recommended routes, and still

[20] Interview SWF2306.
[21] Interview DGF2408.

complete deliveries with high performance scores. In this sense, they "save the day," clearing backlogs that would otherwise overwhelm inexperienced workers.

Because top players routinely perform this invisible corrective labor, they function as a form of semi-skilled labor with exceptionally high replacement costs. This is precisely why, during strikes or moments of intense conflict, algorithmic non-correction, the withholding of the very skills that normally compensate for algorithmic malfunction, becomes a powerful disruptive tactic. This mechanism is best demonstrated when comparing two most recent strikes in Shanwei. Zhao, a top player who participated in both the 2020 and 2023 strikes, described how the 2020 strike was resolved relatively quickly with less damage inflicted because management actively appealed to top players to "rescue orders." The station manager personally begged team leaders and top players to keep delivering, asking them to do so "not because I am the manager, but it's like doing your brother a favor"[22]. Zhao continued to deliver all day, correcting the algorithm malfunction when negotiations were in place. In 2023, however, a newly installed confrontational manager refused to negotiate with even top players. As Zhao recounted:

> That manager didn't even bother talking to us. On the day of the strike, we had worked all day and got exhausted. So, some of us asked for a break ..... Guess what? He rejected it and asked us to recharge our battery at the workplace and eat leftover food if we got hungry – those meals are delivery orders returned by the customers, food had been abandoned for three hours! How could he ask us to eat those cold meals? I was so furious and stop delivering for him – good luck counting on replacement workers."[23]

[22] Interview, SWF2310.
[23] Interview, SWF2310.

Once top players withheld labor, algorithmic malfunctions cascaded unabated. Despite the fact that hundreds of replacement workers were imported from nearby cities, they lacked both local knowledge and algorithmic literacy needed to correct and override the algorithm. As a result, they failed to navigate the terrain when the routing algorithm failed. Most of them completed only one or two tasks, collected minimum bonuses, and passed the remaining hours playing video games in local bars. Without top players to clear backlogs, the strike lasted twelve days and inflicted heavy financial losses on management, eventually compelling the notoriously confrontational managers to stop the proposed pay cut[24].

This tactic resembles the classic work-to-rule strikes (Gary-Bobo and Jaaidane 2014), in that workers withdraw the discretionary, invisible, and voluntary labor on which the system tacitly depends. But algorithmic non-correction diverges in one crucial respect: whereas work-to-rule requires workers to continue working while withholding goodwill, top players in this case deploy algorithmic non-correction by not working at all during strikes. In the form of everyday resistance, however, especially when workers resent a station manager but lack the power to strike, algorithmic non-correction operates as a digital analog to the work-to-rule strike.

Seeing Like an Algorithm: Top Players and the Accumulation of Algorithmic Literacy

Crucially, the capacity to "game the algorithm" is not evenly distributed: it depends on accumulated algorithmic literacy, which is concentrated among top players. This concentration helps explain why top players often occupy strategic positions in collective action. The 2018 Dongguan food-delivery strike illustrates this dynamic. When top players collectively withheld labor in protest against wage cuts and an abusive station manager, orders piled up, and platform

[24] Interview, SWF2314; SWF2306.

operations quickly descended into chaos. The disruption was so severe that the station manager was removed on the second day to placate workers. Management could not stabilize operations until some top players returned. Their ability to exacerbate or alleviate algorithmic malfunctions, combined with the high cost of their replacement, endowed them with real bargaining leverage.

Managers who recognize this influence often cultivate special relationships with top players, treating them differently from rank-and-file workers to keep them loyal. One such top player, Ping, consistently ranked among the top couriers in his district, shared how he quietly received a disputed holiday bonus that many others never got. After Meituan withheld bonuses from dozens of workers who had completed the required shifts, many workers collectively escalated their grievances: they petitioned the company, filed complaints with the labor bureau, and one worker even climbed a tree outside the manager's office and threatened to jump—an extreme act of labor protest meant to signal desperation and force a response. All of these efforts failed. Yet Ping and several other top players quietly received their bonuses. "They wouldn't dare mess with me," Ping explained [25], highlighting how algorithmic literacy and strategic indispensability yielded informal privileges and selective favors.

Top players are typically long-term gig workers who consistently rank near the top of performance metrics and have a detailed understanding of how the algorithm operates. Importantly, they are not necessarily the ones who work the longest hours or simply take the most orders. Rather, they "beat the game" by working smarter: knowing where the hot zones are, how to sequence tasks, and how to exploit algorithmic roles to earn more with less effort, instead of relying on brute force. The making of a top player requires sustained interaction with the algorithm, or, more commonly,

[25] Interview, DGF2406.

integration into peer networks where such knowledge of gaming the algorithm is actively cultivated and shared. Many workers first meet through waiting for orders in the same zone or become teammates in the same station — commonly found in the "service-based platforms" in the Chinese platform economy, where platform companies appoint on-site managers and team leaders in exchange for a more stable and efficient workforce (see, Lei 2021). Ngai, a food-delivery worker, described how his work experience was transformed from the "rookie" status of "not knowing even how best to pick up, cancel, or deliver orders" after joining such a group and being coached by a top player "bit by bit"[26]. It turns out that what needs to be "taught" is the mindset of "seeing like an algorithm." For example, in order to deploy the tactics of algorithmic non-correction, top players must learn to re-examine every micro-decision they made in their everyday labor process. They need to be very strategic about time management, which route to take, whether a detour is worth it, and internalize the algorithm logic in order to outperform it. As a top player explained in front of a group of coworkers:

> In this business, thirty seconds can mean a difference as big as several yuan, you know? You have to think about whether something is worth it. You can't afford to wait at every traffic light; each one wastes a minute or two, so you need to find routes that avoid them… Top players calculate their time very strictly. You need to know how fast you ride—maybe it takes ten seconds to cross a bridge, but you have to factor in whether there's a traffic light ahead, how long the wait time is, whether you should bypass it, how long the customer takes to answer the phone… You must remember clearly which way to go, because one wrong turn will take you overtime.[27]

[26] Interview, DGF2405.
[27] Interview, DGF2408.

Alongside this technical skill, top players also curate and circulate a "hidden transcript" (Scott 1985) about managers and the platform. In everyday gatherings, one of the most common topics is not just wages or bonuses but stories about managers, who accepted bribes, who used the money extracted from gig workers to pursue women at bars, or how companies kept pushing wages downward. During interviews, top players frequently shared anecdotal stories that contrast "good" former managers with the morally corrupted current ones, narratives about how platform companies allegedly shirk their responsibility in car accidents that lead to severe injuries despite having proof of insurance[28], and even stories that describe an unnamed gig worker as a "real hero" who had been blacklisted because of protesting platform companies[29]. These moral stories thereby justifying the use of algorithm-gaming tactics as either everyday resistance or collective disruption.

Through these dense networks of trusted peers, top players become gatekeepers of both tactical repertoires and moral narratives. The hidden transcript they maintain includes not only detailed know-how for gaming and correcting the algorithm, but also a shared moral economy about when and why such tactics are legitimate. During periods of heightened contention, these networks allow top players to convert dispersed, everyday practices into coordinated strategies of collective action, turning algorithmic literacy into bargaining leverage.

[28] Interview, SWF2303; SWR2401.
[29] Interview, DGF2301.

Conclusion and Discussion

Not only gig workers but almost everyone now lives in an "ordinal society" (Fourcade and Healy 2024), in which everyday life is continuously classified, ranked, and reorganized by algorithms. This paper has examined how gig workers respond to such conditions through the lens of symbolic interactionism to understand algorithmic management in the platform economy (Blumer 1986). Workers' conduct cannot be interpreted as a direct effect of the platform algorithm alone; it is produced through ongoing processes of interpretation, experimentation, and negotiation, as workers collectively make sense of the algorithm and the "rules of the game." The impact of technological change is therefore filtered through workers' situated understandings of what the platform algorithm is doing and what can be done in response. This perspective brings worker agency back into the analysis without losing sight of structural constraints. While existing studies rightly warn that everyday efforts to manipulate algorithms may reproduce platform domination by giving workers a misleading sense of autonomy and drawing them deeper into self-exploitation (Cameron 2024; Cini 2025), their collective deployment during strikes carries a different political meaning. In these moments, gaming the algorithm becomes a challenge to the platform power, as workers convert skills learned under algorithmic control into disruptive capacity and bargaining leverage.

Given the far-reaching implications of gaming the algorithm, a related question is whether platform management is aware of these tactical innovations and, if so, whether such awareness ultimately undermines workers' advantages. My research suggests that managers learn about these tactics quickly and actively move to close the loopholes. Technicians continually refine geolocation tracking, tighten time thresholds, and adjust performance metrics to prevent workers

from gaming the system. Indeed, by the time this paper is written, many of these tactics are already well known to management in both cities and can no longer be considered "innovative." Yet, in the context of a strike, managerial adaptations that nerualize workers tactics are costly and imperfect. For example, when the manager in Shanwei manually pinned workers' geolocation to prevent "idling", he also prevented workers from taking orders even when they wanted to join the replacement workforce, generating new inefficiencies and losses. Tactical interaction is thus an ongoing, iterative process (McAdam 1983): gaming the algorithm afford gig workers only temporary bargaining leverage and they have to develop innovative tactics repeatedly.

Several limitations shape the scope of this study. First, the analysis draws on ethnographic and interview data from two cities in Guangdong province and focuses on two major platforms. This raises reasonable questions about generalizability. Despite significant differences in local economies and demographics, I find similar patterns of tactical innovation in gaming the algorithms in both sites and sectors. This is mainly because the basic design of platform algorithms, such as real-time monitoring, performance ranking, and gamified incentives, is highly standardized across the industry. Therefore, I argue that the underlying mechanisms would likely travel beyond my field site with similar algorithmic set-ups: where there is an "algorithmic game," there will be efforts to game it, both for individual and collective advantages.

Second, readers familiar with platform work in other countries may be struck by the role of human managers and the development of hegemony in the Chinese case. The presence of managers, however, is far from new. Lei (2021), for example, draws an influential distinction between "gig platforms," where managers are largely invisible, and "service-based platforms," where managers and team leaders intervene in algorithmic decisions in the Chinese platform

economy. This might suggest that my argument applies only to the latter. I argue instead that it extends to gig platforms as well. Through the lasted development in Chinese platform economy, the boundary between these two types has been increasingly blurred under hegemonic algorithmic control. Ride-hailing platforms like Didi were formally gig platforms, yet since the pandemic, the company has adopted organizational features typical of service-based platforms, such as appointing team leaders and making station managers more visible. This shift reflects platforms' efforts to cope with intensified competition and capture the efficiency gains of a more organized workforce, rendering managerial layers more prominent even on ostensibly "pure" gig platforms.

Third, the argument developed in this paper may not apply to non-hegemonic platforms that lack gamification, ranking, and a strong emphasis on performance data. The very emergence of top players depends on this gamified setting. Under more despotic forms of algorithmic control, where algorithms function primarily as tools of direct coercion and punishment, workers may still try to game the system, but such everyday resistance is less likely to translate into collective bargaining leverage or durable structural power. The mechanisms identified here, therefore, apply most clearly in contexts where hegemonic algorithmic control is deployed at work.

Despite these limitations, the study makes several contributions to research on AI, work, and collective action under platform capitalism. First, it theorizes how hegemonic algorithmic control not only disciplines workers but also paradoxically facilitates collective action by creating the conditions for its own undoing. By making operations dependent on the discretionary "correction labor" of top players, for instance, platforms unintentionally generate a new source of structural power for workers, a tight labor market in which specific skills become both scarce and indispensable.

Second, the article extends McAdam's (1983) framework of tactical innovation to the digital context by identifying a key cognitive mechanism: everyday resistance allows challengers to learn their opponents' structural vulnerabilities and test effective tactics. For gig workers, this means accumulating algorithmic literacy within the labor process itself. When deployed collectively, everyday workarounds can be converted into disruptive tactics during open confrontation, echoing Scott's (2016) argument that "scale shift" occurs when "hidden transcripts" are acted out publicly during heightened contention.

Third, the article develops the concept of the top player to capture heterogeneity among platform workers along the dimension of algorithmic literacy. This perspective extends recent work on differential embeddedness in the platform economy (Schor, Tirrell and Vallas 2024), showing how uneven access to algorithmic literacy shapes who can initiate, sustain, and benefit from collective action. Finally, the analysis revisiting the metaphor of gamification of work and the labor process theory (Burawoy 1979). It demonstrates how "game tactics" are repurposed in platform work, and identifies the conditions under which gamification backfires. By drawing on rarely observed cases where algorithmic resistance "scales up", the paper advances a new framework for understanding collective action under platform capitalism. It highlights a novel form of structural power rooted in gig workers' everyday labor process. Future research that further examines this power across time, place, and platform types can deepen our understanding of how workers organize, negotiate, and resist oppression, as well as the dynamics of labor movements in the digital era.

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